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Mastering Spatial Graph Prediction of Road Networks

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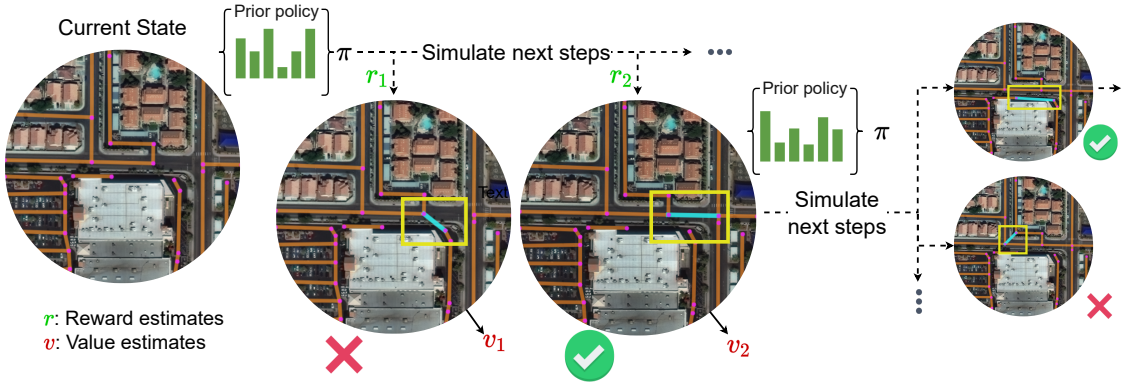


Figure 1. Our agent interacts with the currently generated spatial graph by proposing new edges to be added. A tree-based search produces a sequence of actions that maximizes a reward function based on complex geometrics priors.

Abstract

Accurately predicting road networks from satellite images requires a global understanding of the network topology. We propose to capture such high-level information by introducing a graph-based framework that given a partially generated graph, sequentially adds new edges. To deal with misalignment between the model predictions and the intended purpose, and to optimize over complex, non-continuous metrics of interest, we adopt a reinforcement learning (RL) approach that nominates modifications that maximize a cumulative reward. As opposed to standard supervised techniques that tend to be more restricted to commonly used surrogate losses, our framework yields more power and flexibility to encode problem-dependent knowledge. Empirical results on several benchmark datasets demonstrate enhanced performance and increased high-level reasoning about the graph topology when using a tree-based search. We further demonstrate the superiority of our approach in handling examples with substantial occlusion and additionally provide evidence that our predictions better match the statistical properties of the ground dataset.

1. Introduction

Road layout modelling from satellite images constitutes an important task of remote sensing, with applications in everyday life, such as traffic flow prediction and navigation. The vast amounts of data available from the commercialization of geospatial data, in addition to the need for accurately establishing the connectivity of roads in remote areas, have led to an increased interest in the precise representation of existing road networks. By nature, these applications require structured data types that provide efficient representations to encode geometry, in this case, graphs, a de facto choice in domains such as computer graphics, virtual reality, gaming, and the film industry. These structured-graph representations are also commonly used to label recent road network datasets [60] and map repositories [40]. When dealing with complex predicted structures, however, how well the model optimizes a surrogate objective is not always the best indication of how well the model's predictions are aligned with the required task risk, i.e. the targeted final utilization.

Problems of this nature have been recently extensively studied in the regime of natural language processing, where the misalignment between training objectives for tasks such as machine translation and summarization has garnered considerable attention among the academic community, who have devoted significant efforts to its analysis and elucidation. A popular emerging technique is to first learn to imitate some example outputs and then fix any misalignment issues by using reinforcement learning to adjust the predictions based on a reward function. The reward usually comes in the form of external human evaluation [41, 4].

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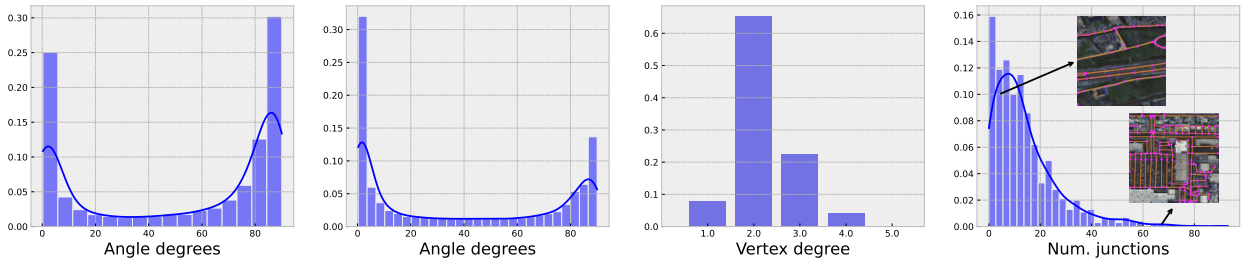


Figure 2. Typical road network features (SpaceNet dataset). From left to right: (a) the distribution of angles between road segments leading to the same intersection is biased towards 0 and 90 degrees, i.e. parallel or perpendicular roads. The same also holds (b) for angles between random road pieces within a ground distance of 400 meters. (c) Most road vertices belong to a single road piece, with a degree of 2. (d) The average number of intersections for areas of 400×400 meters by ground distance.

In this work we manifest how RL fine-tuning can greatly benefit and boost performance when producing intricate geometric predictions. For the task of road layout detection, existing methods, in contrast, mostly rely on pixel-based segmentation models that are trained on masks produced by rasterizing ground truth graphs. Performing pixel-wise segmentation, though, ignores structural features and geometric constraints inherent to the problem. As a result, minimum differences in the pixel-level output domain can have significant consequences in the proposed graph, in terms of connectivity and path distances, as manifested by the often fragmented outputs obtained after running inference on these models. In order to address these significant drawbacks, we propose a new paradigm where we (i) directly generate outputs as spatial graphs and (ii) formalize the problem as a game where we sequentially construct the output by adding edges between key points. These key points can in principle come from any off-the-shelf detector that identifies road pieces with sufficient accuracy. Our generation process avoids having to resort to cumbersome post-processing steps [8, 32] or optimize some surrogate objectives [30, 34] whose relation to the desired qualities of the final prediction can be disputed. Concurrently, the sequential decision-making strategy we propose enables us to focus interactively on different parts of the image, introducing the notion of a current state and producing reward estimates for a succession of actions. In essence, our method can be considered as a generalization of previous refinement techniques [8, 26] with three major advantages: (i) removal of the requirement for greedy decoding, (ii) ability to attend globally to the current prediction and selectively target parts of the image, and (iii) capacity to train based on demanding task-specific metrics.

More precisely, our contributions are the following:

- We propose a novel generic strategy for tuning autoregressive models that removes the requirement of decoding according to a pre-defined order and refines initial sampling probabilities via a tree search.
- We create a synthetic benchmark dataset of pixel-level accurate labels of overhead satellite images for the task of road network extraction. This gives us the ability to simulate complex scenarios with occluded regions, allowing us to demonstrate improved robustness.
- We confirm the wide applicability of our approach by improving the performance of existing methods on the popular SpaceNet and DeepGlobe datasets.

2. Related work

Initial attempts to extract road networks mainly revolved around handcrafted features and stochastic geometric models of roads [6]. Road layouts have specific characteristics regarding radiometry and topology e.g. particular junction distribution, certain general orientation, and curvature (see Fig. 2), that enable their detection even in cases with significant occlusion and uncertainty [19]. Modern approaches mostly formulate the road extraction task as a segmentation prediction task [27, 31, 3] by applying models such as Hourglass [37] or LinkNet [10]. This interpretation has significant drawbacks when evaluated against structural losses because of discontinuities in the predicted masks. Such shortcomings have been addressed by applying some additional post-processing steps, such as high-order conditional random fields [38, 65] or by training additional models that refine these initial predictions [29, 8]. Other common techniques include the optimization of an ensemble of losses. [13] rely on a directional loss and use non-maximal suppression as a thinning layer, while [8] calculate orientations of road segments. Although such auxiliary losses somewhat improve the output consistency, the fundamental issue of producing predictions in the pixel space persists. It remains impossible to overcome naturally occurring road structures, e.g. crossings of roads in different elevations, see Fig. 3.

Previous failure cases have led to more intuitive conceptualizations of the task. Roadtracer [7], iteratively builds a road network, similar to a depth-first search approach, while [13] learn a generative model for road layouts and then apply it as a prior on top of a segmentation prediction mask. Proposed graph-based approaches encode the road

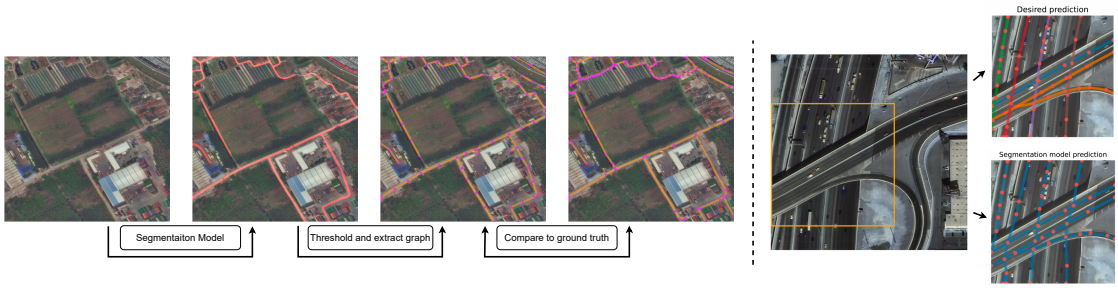


Figure 3. (Left) Typical segmentation-based methods generate an output graph by thresholding a segmentation mask, which can however often lead to fragmented outputs. Predicting segmentation masks also makes it impossible (right) to capture complex road interactions, such as overlapping roads at different elevations.

network directly as a graph, but either operate based on a constrained step-size [56] to generate new vertices or operate on a single step [18, 5], involving use-defined thresholding to post-process the final predictions. Most similar to our work, [26] predict locations of key points and define a specific order traversing them, as also done in [69]. Such autoregressive models have been recently successfully applied with the use of transformers [61] in a range of applications [36, 42, 43, 69] to model constraints between elements, where their supervised training explicitly requires tokens to be processed in a specific order. This specific order combined with the fact that only a surrogate training objective is used, introduces limitations, discussed further in the next section. In order to eliminate this order requirement and to optimize based on the desired metric, while attending globally to the currently generated graph, we propose to use RL as a suitable technique to tune these models.

RL has found success in the past in computer vision applications, we refer the interested reader to [24] for a comprehensive review. These approaches mainly use RL as an auxiliary unit with the goal of improving efficiency [68] or robustness [45]. We differ from these as we use RL to tune pre-trained models with the goal of aligning with a specified task-specific reward. Such tuning was concurrently shown to significantly boost performance for tasks such as object detection, among others [54].

3. Methodology

We parametrize a road network as a graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$ with each vertex $v_i = [x_i, y_i]^\top \in \mathcal{V}$ representing a key point on the road surface. The set of edges $(v_i, v_j) \in \mathcal{E}$, corresponds to road segments connecting these key points. We can then generate a probability distribution over roads by following a two-step process: i) generation of a set of vertices and ii) generation of a set of edges connecting them. Formally, for an image $\mathcal{I}$, a road network $\mathcal{R}$ is derived as:

$$\mathcal{R} = \arg\max_{\mathcal{V},\mathcal{E}} P(\mathcal{V}, \mathcal{E} \mid \mathcal{I}) = P(\mathcal{E} \mid \mathcal{V}, \mathcal{I}) P(\mathcal{V} \mid \mathcal{I}). \quad (1)$$

The graph nodes typically correspond to local information in an image, and we therefore resort to a CNN-based model to extract key points, providing the set $\mathcal{V}'$, that sufficiently captures the information in the ground truth graph $\mathcal{G}$. The construction of edges, however, requires higher-level reasoning that can cope with parallel roads, junctions, occlusions, or poor image resolution, among other difficulties.

Considering probabilistic models over sequences and using the chain rule, we can factorize the joint distribution as the product of a series of conditional distributions

$$P(\mathcal{E} \mid \mathcal{V}, \mathcal{I}; \sigma) = \prod_{n=1}^{N_\mathcal{E}} P(e_{\sigma(n)} \mid e_{<\sigma(n)}, \mathcal{V}, \mathcal{I}), \quad (2)$$

where $e_{<\sigma(n)}$ represents $e_{\sigma(1)}, e_{\sigma(2)}, \ldots, e_{\sigma(n-1)}$ and $\sigma \in S_{N_\mathcal{E}}$ denotes the set of all permutations of the integers $1, 2, \ldots, N_\mathcal{E}$, with $N_\mathcal{E}$ the number of edges. For our work we consider the setting where these sequences are upper bounded in length, i.e. $N_\mathcal{E} \leq N_{\max}$, since we are dealing with satellite images of fixed size. Autoregressive models (ARMs) have been used to solve similar tasks in the past by defining a fixed order of decoding [39, 59, 36, 42]. In our case, this would correspond to sorting all key points by their coordinates x and y and generating edges for each of them sequentially. We call this the *autoregressive order*. There are, however, two major drawbacks.

First, the evaluation metrics used for this task define a buffer region in which nodes in the ground truth and the predicted graph are considered to be a match. Therefore, a newly generated edge can be only partially correct when only partially overlapping with the ground truth graph. This non-smooth feedback comes in clear contrast to the supervised training scheme of ARMs, minimization of the negative log-likelihood, that assumes perfect information regarding the key points' locations, i.e. that the sets $\mathcal{V}$ and $\mathcal{V}'$ are the same. In practice, this condition is rarely met as the exact spatial graph can be represented in arbitrarily many ways by subdividing long edges into smaller ones or due to small perturbation to key points' locations. It is thus imperative that our model can estimate the expected improvement of adding selected edges, which implicitly can also signal when to appropriately end the generation process.

Second, the requirement to decode according to the autoregressive order introduces a bias and limits the expressiveness of the model [58]. As a result, it can lead to failures in cases with blurry inputs or occlusions [26]. We employ RL to overcome these deficiencies and align the model with the required objective. In more detail, we tune an autoregressive model, described in Section 3.1, with the ability to search into sequences of actions in the future, as described in Section 3.2. Our novel generic strategy improves the autoregressive model without requiring significantly more computational resources.

3.1. Autoregressive model

We start by introducing a base autoregressive model, illustrated in Fig. 4. Given an image and a set of key points, i.e a set of vertices $\mathcal{V}'$, our model produces a graph by sequentially predicting a list of indices corresponding to the graph's flattened, unweighted edge-list. Each forward pass produces probabilities over the set of key points, which leads to a new index after sampling. A successive pair of indices defines an edge as its two endpoints. A special end-of-sequence token is reserved to designate the end of the generation process.

Following [63, 53], we begin by extracting visual features per key point, by interpolating intermediate layers of a ResNet backbone to the key points' locations, which are further augmented by position encodings of their locations. We then further process these features using two Transformer modules. The first transformer (Transformer I in Fig. 4) encodes the raw features of the key points as embeddings. The second transformer (Transformer II in Fig. 4) takes as input the currently generated edge list sequence, corresponding to the currently partially generated graph. Edges are directly mapped to the embeddings of their comprising key points, supplemented by position and type embeddings, to differentiate between them, as shown in Fig. 5 (a). An additional global image embedding also extracted by the ResNet is used to initialize the sequence. The Transformer II module produces a single hidden state, which is linked with the $N_{\mathcal{V}'} + 1$ (corresponding to the provided key points, supplemented by the special end of the generation token) key points' embeddings by a pointer network [62], via a dot-product. This allows the generation of a probability distribution of variable length depending on the current set $\mathcal{V}'$, instead of using a fixed action space.

3.2. Augmented search

In order to address the problems of greedy decoding (analyzed in Section 3), we frame our road extraction task as a classical Markov-decision process (MDP). The generation of a graph for every image defines an environment, where the length of the currently generated edge list determines the current step. Let $\boldsymbol{o}_t$, α_t and r_t correspond to the observation, the action, i.e. the selected key point index, and the observed reward respectively, at time step t. The aim is to search for a policy that maximizes the expected cumulative reward over a horizon T, i.e., $\max_\pi J(\pi) \coloneqq \mathbb{E}_\pi[\sum_{t=0}^{T-1} \gamma^t r_t]$ where $\gamma \in (0, 1]$ indicates the discount factor and the expectation is with respect to the randomness in the policy and the transition dynamics. We set the discount factor to 1 due to the bounded time horizon.

Instead of training a reward model [41], we employ established graph-theoretic metrics that also allow the definition of intermediate rewards, accelerating initial training. More formally, each action leads to the selection of a new key point, with new edges being added once every two actions. The addition of a new edge leads to a revision of the predicted graph and triggers an intermediate reward

$$r_t = \text{sc}(\mathcal{G}_{\text{gt}}, \mathcal{G}_{\text{pred}_t}) - \text{sc}(\mathcal{G}_{\text{gt}}, \mathcal{G}_{\text{pred}_{t-1}}), \tag{3}$$

where $\text{sc}(\mathcal{G}_{\text{gt}}, \mathcal{G}_{\text{pred}_t})$ is a similarity score between the ground truth graph $\mathcal{G}_{\text{gt}}$ and the current estimate $\mathcal{G}_{\text{pred}_t}$. Discussion of the specific similarity scores used in practice is postponed for Section 3.3.

A proper spatial graph generation entails (i) correct topology and (ii) accurate location prediction of individual roads. For the latter, intermediate vertices of degree 2 are essential. We call a road segment (RS), an ordered collection of edges, between vertices of degree $d(.)$ two (or a collection of edges forming a circle):

$$\begin{aligned}
&\text{RS} = \{(\boldsymbol{v}_{\text{rs}_1}, \boldsymbol{v}_{\text{rs}_2}), \ldots, (\boldsymbol{v}_{\text{rs}_{k-1}}, \boldsymbol{v}_{\text{rs}_k})\} \\
&\text{s.t } (\boldsymbol{v}_{\text{rs}_i}, \boldsymbol{v}_{\text{rs}_{i+1}}) \in \mathcal{E} \text{ for } i = 1, \ldots, k-1 \\
&d(\boldsymbol{v}_{\text{rs}_i}) = 2, \text{ for } i = 2, \ldots k-1, \\
&(d(\boldsymbol{v}_{\text{rs}_1}) \neq 2 \text{ and } d(\boldsymbol{v}_{\text{rs}_k}) \neq 2 \text{ or } \boldsymbol{v}_{\text{rs}_1} = \boldsymbol{v}_{\text{rs}_k}).
\end{aligned}$$

During the progression of an episode (i.e. the sequential generation of a graph), the topological nature of the similarity scores in Eq. 3 implies that the effect of each new edge to the reward will be reflected mostly once its whole corresponding road segment has been generated. To resolve the ambiguity in the credit assignment and allow our agent to look ahead into sequences of actions, we rely on Monte Carlo Tree Search (MCTS) to simulate entire sequences of actions. We use a state-of-the-art search-based agent, MuZero [48], that constructs a learnable model of the environment dynamics, simulating transitions in this latent representation and leading to significant computational benefits. At every step, the enhanced model with its new components produces value [50] and reward estimates, which define an exploration strategy. Simulations into future sequences of actions are performed based on this strategy, helping the model to make decisions based on future outcomes, not yet experienced. Compared to a simpler PPO objective [49, 41], that in our experience did not lead to significant improvements, MCTS generates a series of simulations traversing the tree from a root node, generating more

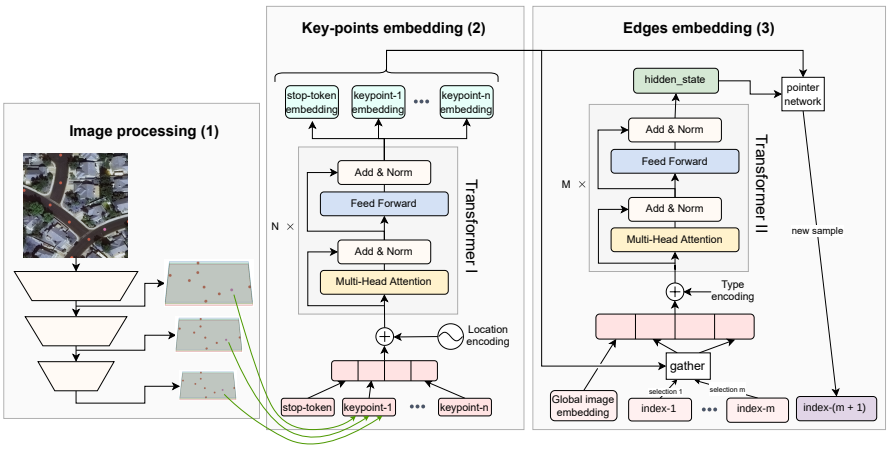


Figure 4. The autoregressive model with its three main components. (1) A backbone image model (ResNet) that extracts features for each key point at different scales, along with a global image embedding. (2) A key point model embeds visual and location features of distinct key points. (3) An edge embedding model relates the current edge sequence with the respective key points. Each edge token (signalled with the tokens 'index-i') corresponds to an index specifying the respective key point. A pair of such tokens designates an edge as its two endpoints. At the end of (3) we obtain a new distribution over key points that leads to an incremental update to the graph, after sampling.

stable value and reward estimates. MuZero requires three distinct parts (see also Fig. 5):

1. A representation function f creates a latent vector of the current state $\boldsymbol{h}_t = f_\theta(\boldsymbol{o}_t)$, in our case the autoregressive model, shown in Fig. 4. Our current latent representation $\boldsymbol{h}_t$ contains the graph's hidden state along with the key points' embeddings used to map actions to latent vectors. As key points remain the same throughout the episode, image-based features (Components (1) and (2) in Fig. 4) are only computed once.

2. A dynamics network g, we use a simple LSTM [20], as commonly done, that predicts the effect of a new action by predicting the next hidden state and the expected reward: $(\hat{\boldsymbol{h}}_t, \hat{r}_t) = g_\theta(\tilde{\boldsymbol{h}}_{t-1}, \alpha_t)$. We can replace $\tilde{\boldsymbol{h}}_{t-1}$ with the latent representation $\boldsymbol{h}_{t-1}$, or its previous computed approximation $\hat{\boldsymbol{h}}_{t-1}$ for tree search of depth larger than 1.

3. A prediction network ψ, that estimates the policy and the value for the current state $(\boldsymbol{p}_{t+1}, v_t) = \psi_\theta(\tilde{\boldsymbol{h}}_t)$. We compute the policy via a pointer network as described in Section 3.1. Value estimates are produced by a simple multi-layer perceptron.

The dynamics network guides the search and evaluates the expected reward of actions. For every newly generated edge, we also explicitly inform the network regarding the creation of new intersections and the expected relative change in the overall road surface generated via embeddings (see Fig. 5). By using the dynamics network, we bypass the expensive call to the decoder module during the search, and can instead approximate small modifications in the latent representation directly. For our experiments, the dynamics

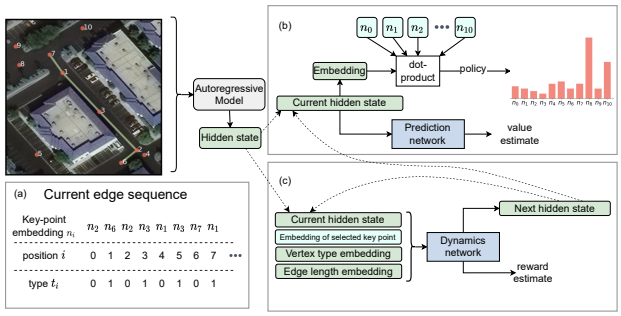


Figure 5. The autoregressive model generates a hidden state corresponding to the graph embedding. (a) When doing so, the edge decoder directly attends to the input key points' embeddings, augmented by position and type embeddings. (b) The prediction network uses the hidden state to produce value and policy predictions. A pointer network allows an intuitive scale of the action space by the number of key points. (c) The dynamics network simulates trajectories by estimating new hidden states and rewards. For newly generated edges, it takes as input the embeddings of the new key points, but also the degree of the two vertices involved and the length of the newly proposed generated edge.

network requires up to 90 times less floating-point operations to simulate trajectories, compared to using the edge embeddings' decoder. Effectively, our method does not involve significantly more computation budget compared to the base autoregressive model. More details regarding the MuZero training and exploration are provided in Section 4 and in the appendix.

3.3. Evaluation metrics

We adopt the same evaluation metrics both as a comparison between different methods and to determine the reward for our agent, through Eq. 3. We use the relaxed versions of

precision, recall, and intersection over union for pixel-level predictions *Correctness/Completeness/Quality* (CCQ) [66, 64]. As graph-theoretic metrics we use *APLS* [60] and additionally include new metrics introduced in [14] that compare *Paths*, *Junctions* and *Sub-graphs* of the graphs in question, producing respectively *precision*, *recall* and f_1 scores. More details can be found in the appendix.

4. Experiments

Implementation details. We resize images to 300×300 pixels, standardizing according to the training set statistics. For exploration, we initialize workers using Ray [33] that execute episodes in the environment. For training, we unroll the dynamics function for $t_d = 5$ steps and use priority weights for the episode according to the differences between predicted and target values. Our algorithm can be considered as an approximate on-policy TD(λ) [55] due to the relatively small replay buffer. We reanalyse older games [48] to provide fresher target estimates. Unvisited graph nodes are selected based on an upper confidence score, balancing exploration with exploitation, similar to [51]. We add exploration noise as Dirichlet noise and select actions based on a temperature-controlled sampling procedure, whose temperature is reduced during training.

Given the limited high-quality available ground truth labels [52] and to accelerate training, we employ modifications introduced in EfficientZero [71]. We investigate adding supervision to the environment model and better initialize Q-value predictions similar to the implementation of Elf OpenGo [57]. We further scale values and rewards using an invertible transform inspired by [44]. Here, we predict support, as fully connected networks are biased towards learning low-frequency representations [21]. Selecting new actions involves generating simulations that can be done expeditiously given the small dimension of the latent space and the modest size of the dynamics network. Finally, to generate key points, we skeletonize segmentation masks provided by any baseline segmentation model, by thresholding the respective segmentation masks produced and applying RDP-simplification [17, 46]. Selecting an appropriate threshold and subdividing larger edges guarantees that the generated set $\mathcal{V}'$ adequately captures most of the ground truth road network, leaving the complexity of the problem for our model to handle. We will use the term *ARM* to denote the autoregressive model and *Ours* for the RL-tuned model.

4.1. Synthetic dataset

Initially, a simplified setting is chosen wherein the complete governance of both the nature and intensity of the difficulty inherent in the road-extraction task can be regulated. We generate a dataset of overhead satellite images of a synthetic town using CityEngine[1]. We randomly specify vegetation of varying height and width along the side walks of the generated streets, leading inadvertently to occlusions of varying difficulty. The simulated environment allows specifying *pixel-perfect masks* regarding both *roads and trees* occluding the road surface based on the provided camera parameters [23]. We defer more details regarding the generation process and dataset examples to the supplementary material.

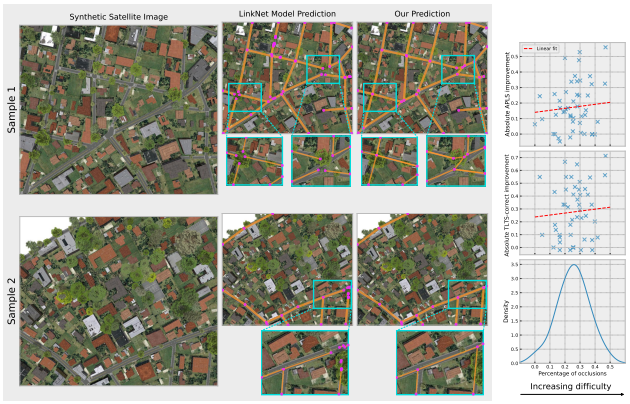


Figure 6. (Left) Examples of fragmented outputs from the LinkNet model under cases of significant occlusion and how our approach performs in these demanding circumstances. (Right) The performance gap between our method and the same baseline is wide, based on topological spatial graph metrics, for a wide range of difficulties in the images. More details and examples are given in the supplementary material.

We compare our method by training on our dataset a *LinkNet* model [10], a popular segmentation model that has been widely used in the remote sensing community [25]. Even in this synthetic and thus less diverse scenario, the deficiency of segmentation models to rely mostly on local information, with no explicit ability for longer-range interactions, is evident. Fig. 6 illustrates examples of such oversegmented predictions and how our approach can improve on them. We also define a 'difficulty' attribute per synthetic satellite image, quantifying the occlusions as a percentage of the ground truth road mask covered. We observe a considerable absolute improvement in topological metric scores when training our model on this synthetic dataset, compared to the baseline, for varying image difficulty.

4.2. Real datasets

We assess our approach on the SpaceNet and DeepGlobe datasets. A single image is provided to our method, with a spatial graph being generated as the output. We use the same train-test splits as in [8] to promote reproducibility, while results are reported for the final combined graph on the original image scale. No pre-training on the synthetic

[1] https://www.esri.com/en-us/arcgis/products/arcgis-cityengine/overview

	Method	CCQ			TLTS		APLS ↑	Path-Based			Junction-Based			Sub-graph-Based
		corr. ↑	comp. ↑	qual. ↑	corr. ↑	2l+2s ↓		pre. ↑	rec. ↑	f1 ↑	pre. ↑	rec. ↑	f1 ↑	f1 ↑
SpaceNet	DeepRoadMapper [29]	0.6943	0.6838	0.5386	0.4110	0.1012	0.5143	0.5958	0.6400	0.6171	0.6293	0.7443	0.6820	0.6783
	Segmentation [28, 22]	0.7493	0.7094	0.5969	0.4143	0.0828	0.5454	0.6909	0.6863	0.6885	0.7186	0.7710	0.7438	0.7117
	LinkNet [10]	0.8100	0.7449	0.6409	0.4894	0.0743	0.5743	0.6719	0.6460	0.6586	0.6985	0.7809	0.7374	0.7576
	Orientation [8]	0.8070	0.8001	0.6862	0.5594	0.0884	0.6315	0.7175	0.7280	0.7227	0.7552	0.7591	0.7571	0.7802
	Sat2Graph [18]**	0.6917	0.7351	0.5734	0.5802	0.1104	0.5951	0.5952	0.5416	0.5671	0.7474	0.5951	0.6626	0.7180
	SPIN road mapper [5]	0.7837	0.7988	0.6621	0.5922	0.1058	0.6422	0.7276	0.7265	0.7270	0.7621	0.7827	0.7722	0.7837
	Ours	0.8150	0.8092	0.6932	0.5970	0.0732	0.6587	0.7383	0.7613	0.7496	0.7845	0.7821	0.7833	0.7948
Deep Globe	LinkNet [10]	0.8012	0.8676	0.7328	0.6640	0.0804	0.6525	0.6882	0.6920	0.6901	0.7675	0.7444	0.7558	0.7879
	Orientation [8]	0.8243	0.8857	0.7545	0.6866	0.1047	0.7012	0.6937	0.8082	0.7465	0.7624	0.7939	0.7778	0.8282
	Ours*	0.8223	0.8979	0.7494	0.7242	0.0743	0.7400	0.7150	0.8274	0.7671	0.7912	0.8283	0.8093	0.8391

* We do not fine-tune our model on the DeepGlobe dataset but instead refine predictions standardizing according to train dataset statistics.

** The authors provided predictions corresponding only to a center crop of the original SpaceNet dataset images. Also, note that the test set is different from the one reported on the rest of the methods, see also the appendix.

Blue: best score, Green: second best score, Gray: results reported in different test set

Table 1. Quantitative results for the SpaceNet and DeepGlobe datasets.

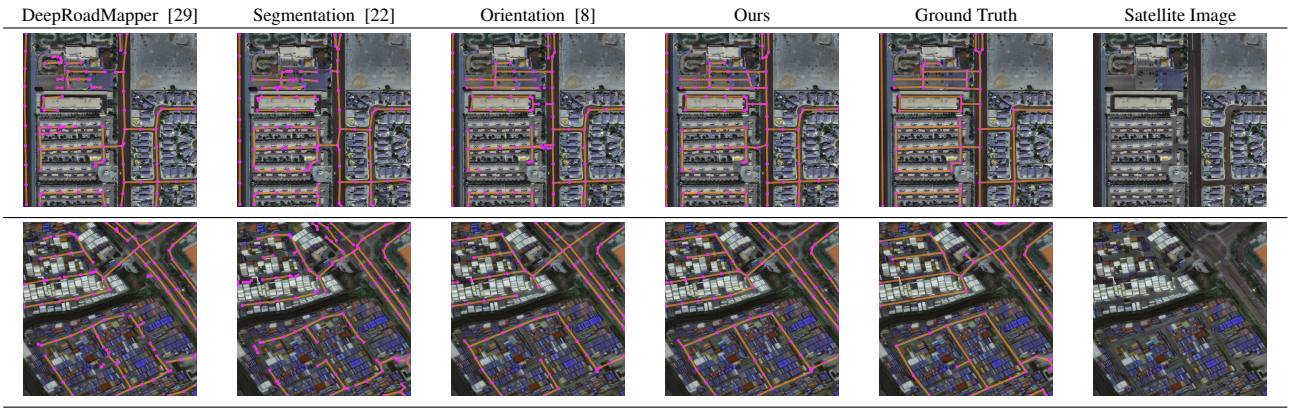


Table 2. Qualitative results of improved connectivity. We recommend zooming in for more details.

Table 3. APLS metric and perplexity (bits of information per edge) results for the SpaceNet dataset with provided key points that adequately capture the ground truth graph. Autoregressive order is defined in Section 3, while random order entails random permutation between edges and the order of key points within edges. To calculate perplexity for our method we use the initial predicted policy distribution, without any additional search. More details on baselines and general approach are presented in the appendix.

Metric \ Method	Random	Cls	GCN	ARM	Ours
APLS	0.008	0.430	0.594	0.894	**0.928**
Bits per edge: Autoregressive order	8.743	-	-	**0.528**	4.321
Bits per edge: Random order	8.743	-	-	28.74	**4.432**

dataset takes place.

4.2.1 Comparison to Baselines

We initially train the ARM model on ground-truth graph information by using a set of key points $\mathcal{V}' = \mathcal{V}$. Even in this scenario, RL tuning increases the semantic quality of the extracted graphs as seen in Table 3. In cases where the set $\mathcal{V}'$ is imperfect, and for reasons elaborated in Section 3, newly proposed edges may be only partially correct. Under such circumstances, supervised training is infeasible, hence we rely on RL to produce the graph that leads to the maximum cumulative reward, i.e. our selected evaluation metrics. We conduct comparative analysis against the following approaches; we explore powerful CNN architectures, by training a *Segmentation* model with a ResNet backbone. We evaluate *DeepRoadMapper* [29], a model that refines previous segmentation by adding connections along possible identified paths. As done by [8] we notice that in complex scenarios, the effect of this post-processing step is not always positive. We also evaluate against *LinkNet* [10], and *Orientation* [8], which is trained to predict simultaneously road surfaces and orientation, and the relevant baselines [18, 5].

Quantitative results in Table 1 and visual inspection in Table 2, affirm that the global context and the gradual generation incite a better understanding of the scene, leading to consistently outperforming topological metric results compared to the baselines. We remark that our predictions are more topologically consistent with fewer shortcomings, such as double roads, fragmented roads, and over-connections. This is further supplemented by comparing the statistics of the predicted spatial graphs in Fig. 7. We further

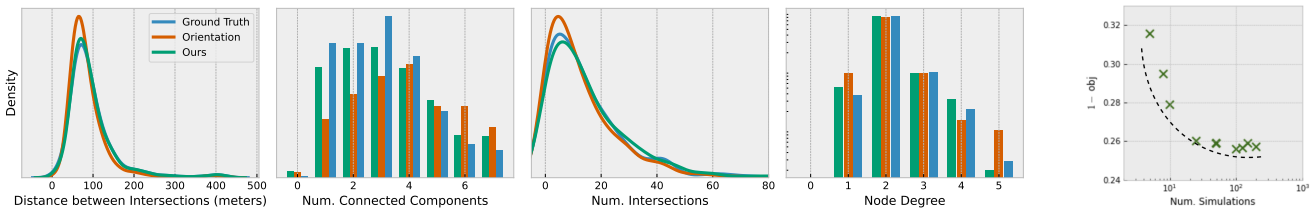


Figure 7. (Left) Comparison of generated graph statistics, following the same post-processing, averaged across regions of 400×400 meters in ground distance. Orientation refers to the method of [8]. (Right) The estimated Pareto front achieved through the evaluation of various runs, where a different number of MCTS simulations is being executed at each state. *obj* refers to an average of the *APLS*, *Path-based f1*, *Junction-based f1* and *Sub-graph-based f1* metrics.

showcase the transferability of our model by employing it with no fine-tuning (apart from dataset-specific image normalization) on the DeepGlobe dataset. We can refine previous predictions by adding missing edges, leading to more accurate spatial graph predictions, as shown in Table 1. Our conjecture that road structures and their geometric features exhibit a high degree of recurrence across various locations and geographic regions worldwide is thus confirmed.

4.2.2 Ablation study

We experimented attending to image features for the two transformer modules by extracting per-patch visual features from the conditioning image $H^{\text{img}} = [\boldsymbol{h}_1^{\text{img}}, \boldsymbol{h}_2^{\text{img}}, \dots]$, as done in the Vision Transformer [16]. This did not lead to significant improvements, which we attribute to over-fitting. In Table. 4 we highlight the relative importance of some additional components for the final predictions. As efficiency is also of particular importance to us, we further visualize the effect of varying the simulation depth of the dynamics network during training (Fig. 7 (right)). Surprisingly perhaps, our method performs consistently even with a limited number of simulations.

In the appendix, we provide incremental results for the task of predicting road networks based on an optimal set of key points and insights concerning interpretability and further comparison to baselines based on the varying difficulty of the predicted underlying road networks. We also give more information regarding the generation of the synthetic dataset and the model architecture. Finally, we provide more implementation decisions, including details on exactly how key points and generated and how individual patch-level predictions are fused together. It is important to emphasize that our proposed method can operate effectively even when working with partially initialized predictions, making it a practical refinement approach that can be applied to existing baselines. By initializing our model based on the ARM model, we can achieve a rapid fine-tuning phase. Additionally, by utilizing the learned environment model, which avoids the need for expensive calls to the edge embedding model at each simulation step in MCTS, we can train the model even using a single GPU.

Model	APLS	P-f1	J-f1	S-f1
Ours	**0.6587**	**0.7496**	**0.7833**	**0.7948**
– autoregressive pre-training	-15.3%	-13.7%	-14.1%	-13.3%
– visual features for key-points	-13.4%	-12.7%	-12.1%	-12.3%
– tree-search during evaluation	-2.1%	-1.4%	-1.7%	-1.1%
+ cross attend to image features	+0.2%	-0.4%	-0.7%	-0.3%

P: Path-based, J: Junction-based, S: Sub-graph-based

Table 4. Ablations study on SpaceNet dataset.

5. Conclusions

We presented a reinforcement learning framework for tuning autoregressive tasks for the task of generating a graph as a variable-length edge sequence, where a structured-aware decoder selects new edges by simulating action sequences into the future. Importantly, this allows the model to better capture the geometry of the targeted objects. One advantage of the proposed method is that the reward function is based on (non-continuous) metrics that are directly connected to the application in question. Our approach does not require significantly more computational resources compared to state-of-the-art supervised approaches, and in addition, it can be used to refine predictions from another given model. We also remark that the direct prediction of a graph enables the concurrent prediction of meta-information about the edges, including, for instance, the type of road (e.g. highway, primary or secondary street, biking lane).

Our approach opens the door to several directions for future work. For example, we have assumed that a pre-defined model gives the location of key points, but one could instead augment the action space to propose new key points' locations. Other promising directions include the direct prediction of input-dependent graph primitives, e.g. T-junctions or roundabouts. Finally, we emphasize that our approach is suitable for a wide variety of applications where autoregressive models are typically used, which we intend to look into in the future. Such applications include among others, Scene Graph Generation [67, 70] and Visual reasoning/Factual Visual Question Answering [11, 35].

6. Reproducibility Statement

We have taken multiple steps to ensure the reproducibility of the experiments. We refer the reader to the appendix for a complete description of the training protocol. We have also released the code as part of the supplementary material.

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